RadXAI: Probing and Mitigating Hallucination in Radiology VLMs with Visual Retrieval-Augmented Generation and Policy Optimization

Abstract

Visual Language Models (VLMs) have extensive clinical applications resulting in an increasing demand for medical vision-language models (Medical VLMs). Despite the promising results of VLMs, they are susceptible to hallucinations that is clinically dangerous and poorly captured by text generation metrics. Visual retrieval augmented generation (VRAG) is a remedy in reducing hallucinations. In this work, we reproduce and extend VRAG on the CheXpert Plus corpus with LLaVA-1.5-7B as the backbone, and measure hallucination directly through a controlled probing protocol. We develop a framework that takes an X-ray image as input to perform visual question answering (VQA) and medical report generation while mitigating hallucinations in Med VLMs. Experiments are conducted on the CheXpert dataset. The LLaVA model was finetuned on raw-reports of the patients as the report of the patient of the frontal chest X-ray image followed by Visual Retrieval Augmented Generation (VRAG). We automatically constructed a VQA test pair for the radiology data and evaluate three models: a base model, a domain-adapted model, and a VRAG finetuned model. All are evaluted under three retrieval conditions: no retrieval, VRAG retrieval, and a random-retrieval control. On CheXpert Plus, V-RAG finetuning with retrieval raises finding level precision from 0.549 to 0.844 and MCC from $+0.159$ to $+0.472$, corresponding to a reduction in the hallucination rate from 45.1% to 15.6%; the ordering *no-retrieval* $<$ *random* $<$ *VRAG* holds for both trained models on CheXpert Plus, MIMIC-CXR and IU X-Ray datasets. We found that retrieval relevance drives hallucination reduction. To reduce the bias, we further do reasoning chains using Reinforcement Learning with Verifiable Reward (RLVR), enabling the model to do better reasoning to provide reliable and fair answers.

Introduction

Recent advances in Vision Large Language Models (VLLMs) have facilitated the healthcare applications of Artificial Intelligence (AI). This have resulted in advancing research on Medical-VLMs (Med-VLMs) zhao2025medrag, Hartsock2024 models that are pre-trained on medical image and text understanding, such as Biomed CLIP zhang2023biomedclip, MedGemma sellergren2025medgemma, and Med-Flamingo moor2023medflamingo. These are trained on generalist medical data drawn from scientific articles in National Institutes of Health (NIH), PubMed Central (PMC), as well as clinical imaging archives and medical textbook corpora than on task-specific clinical datasets.

However, LLMs trained on generalized corpora often struggle with medical tasks that demand specialized domain knowledge. This limitation stems from the difficulty in capturing subtle clinical nuances. To that end, LLMs often do not reason effectively in complex domain specific scenarios, and precise reasoning requires understanding semantically similar yet clinically distinct entities.

Over time, researchers have focused on improving the reliability of Med-VLMs using various techniques including fine tuning and the use of Retrievel Augmented Generation (RAGs) lewis2020retrieval. Fine tuning adapts a foundation model ready for specific downstream task, but are prone to hallucinations with high confidence, when applied to specific domain applications like healthcare applications. RAG-based methods, by contrast, are used for fetching the information from external databases and augment it with the responses generated by the LLMs. Despite it helps to give external references during the inference stage to give answers based on context, you do have some limitations in itself. These models are prone to hallucinations, which are critical in healthcare with answers not grounded based on the visual inputs. In Mirage asadi2026mirage, the authors came to an striking finding that VLM's produce confident responses even when the input image is entirely absent. In critical medical scenarios, such unsupported and fabricated outputs can result in potentially fatal decisions. Motivated by these challenges, we present the following contributions in our work:

- We present **RadXAI**, grounding Med-VLM outputs in retrieved visual evidence, and measure hallucination through controlled entity probing rather than text-similarity scores.
- We build **test_vqa**, a hallucination benchmark across CheXpert, MIMIC-CXR, and IU X-Ray. It is POPE-style (Li et al., 2023) hallucination benchmark of automatically constructed, per-image-balanced, semantically

sibling-safe Yes/No questions, scored with MCC and standard classification metrics under the three retrieval conditions.

- We do external benchmarking of the same protocol on MIMIC-CXR (Johnson et al., 2019) and IU X-Ray (Demner-Fushman et al., 2016), showing the effect is not confined to the training distribution.
- We conducted a systematic evaluation across subsets of the CheXpert, MIMIC-CXR, and Indiana University chest X ray datasets, using ROUGE, F1, Precision, Recall, and Matthews Correlation Coefficient (MCC) to assess both generalization quality and reliability of the framework.
- RLVR approach that learns *when* to retrieve via GRPO with verifiable reward, to learn whether the retrieval is worth invoking

Literature Review

VLMs for Clinical Intelligence

The remarkable predictive performance of LLMs has given rise to a new class of medical LLMs that are capable of jointly reasoning over clinical images and text. Biomed-CLIP (Zhang et al., 2023) employs contrastive image-text pretraining similar to CLIP (Radford et al., 2021). It is trained on PMC-15M image-text dataset from scientific papers collected from PubMed Central (PMC). Med-Flamingo (Moor et al., 2023) extended OpenFlamingo through continued pretraining on medical publications and textbooks, targeting few-shot visual question answering over interleaved image-text sequences. Recently MedGemma (Sellergren et al., 2025) by Google combined a Gemma-3 based backbone with a domain tuned vision encoder (MedSigLIP), trained across radiology, dermatology, and pathology images. However, these models trained on literature corpora rather than fine tuning to narrow clinical tasks and hence fail to capture semantically overlapping clinical entities.

RAGs for Medical LLMs

The pre-trained LLMs achieves SOTA results on fine tuned downstreamed tasks, however still is challenging in specific knowledge intensive tasks like healthcare applications. Retrieval Augmented Generation (RAGs) by (Lewis et al., 2020) enhance the answer generated by LLMs by incorporating information fetched from an external database or knowledge bases that contains medical documents, reports or EHR (Li et al., 2026; Chen et al., 2025; Li et al., 2025; Zhao et al., 2025). RAGs retrieves the top *k* documents from the knowledge base after finding the similarity between the query embeddings and the document embeddings. However, they have the ability to adapt to complex and evolving medical cases. In the work, (Abo El-Enen, Saad, and Nazmy, 2025) the authors lists the potential challenges, for instance domain specific nuances in Med-LLMs as medical corpora contains specialized terminologies, or contextual dependencies that general-purpose LLMs may misinterpret.

Reinforcement Learning from Human and AI Feedback

In recent years, the reasoning capabilities of LLMs have improved significantly. This enhanced reasoning ability strengthens Retrieval-Augmented Generation (RAG), particularly for queries that require multi-step retrieval and complex reasoning. Reinforcement learning (RL) has emerged as an effective approach for steering model behavior toward more reliable and faithful outputs beyond architectural or RAG-based methods (Chen et al., 2026). RL optimizes model behavior by assigning rewards to desirable responses, thereby improving reasoning and decision-making. Reinforcement Learning from Human Feedback (RLHF) trains a reward model using human preference judgments and subsequently optimizes the policy against this learned reward signal. Reinforcement Learning from AI Feedback (RLAIF) (Lee et al., 2024) was proposed as a scalable alternative, replacing human preference annotations with preferences generated by an LLM acting as a judge. RLAIF has demonstrated performance comparable to RLHF on tasks such as summarization and dialogue generation while substantially reducing the cost of human annotation. More recently, Reinforcement Learning with Verifiable Rewards (RLVR) has gained attention as an alternative paradigm that replaces subjective preference-based rewards with objective, automatically verifiable reward signals. Instead of relying on human or AI preference models, RLVR rewards responses based on whether they satisfy predefined correctness criteria or verifiable outcomes, making it particularly suitable for domains where ground-truth answers or verification mechanisms are available. This objective reward formulation encourages models to produce more accurate and logically consistent reasoning while avoiding biases inherent in preference-based supervision. In domain-specific settings, recent work has begun applying AI-generated and verifiable reward signals to medical answer correctness, suggesting that RLAIF and RLVR can reduce human-feedback bias while guiding models toward more reliable and clinically faithful outputs.

Methology

In this work, our goal is to construct a clinically grounded, Visual Question Answering (VQA) dataset for chest X-rays from CheXpert Plus (Chambon et al., 2024) and for any given a query radiograph x_q, produce a report r_q. The chest X-ray reports into a structured disease entities and binary Visual Question Answering (VQA) pairs and fine-tune a vision language model (LLaVA) for medical visual question answering. The pipeline does clinical entity extraction from raw radiology reports followed by deterministic VQA pair construction from the extracted entities. Then LoRA-based fine-tuning of LLaVA on the resulting dataset to retrieve the data from the VisionRAG (VRAG) and uses reinforcement learning with Group Relative Policy Optimization (GRPO) algorithm. An overview of the approach is as illustrated in Figure 1.

Baseline

The base LLAVA results in hallucinations, which is attributed to the fact that it lacks medical grounding. This is because it was never trained on curated radiological data with clinical supervision. To that end, we use use the Vision Retrieval Augmented Generation (V-RAG) *i.e*, using images in RAGs.

Here, we reproduce and extend V-RAG on the CheXpert Plus corpus with LLaVA 1.5-7B as the backbone and study hallucination directly through a controlled entity probing protocol than through text scoring alone. We adopt a three-stage model progression that incorporates: (i)a base model (*llava*), *(ii)* domain adapted model obtained by supervised finetuning on CheXpert reports (*llava_s*), and *(iii)* a V-RAG model obtained by further finetuning of *llava_s* on a mixture of retrieval-oriented multi-image tasks (*llava_vrag*). All these are evaluated under three retrieval conditions: no retrieval, V-RAG retrieval, and a random control retrieval. The random control is essential, as it separates the benefit of clinically relevant evidence from the benefit of merely supplying additional context.

Data Preprocessing and Preperation

As the first step, we make a reproducible dataset preparation pipeline where the input is the raw written radiology report by doctors, and the output is a structured set of positive and negative VQA samples. The loaded raw reports are filtered for the frontal images only, then entity is extracted and normalized. Entity extraction is done using Qwen2.5-14B (Instruct-bnb-4bit) (Ahmed et al., 2025), with a 16,384 token context window and continued with a Positive-rich threshold for more than two positive entities.

The CheXpert Plus dataset comprises 223,462 unique pairs of radiology reports and chest radiographs in 187,711 studies of 64,725 patients. The image is used instead of the DICOM and that contributes to the image-text pair. We formulated a structured metadata and stable identifiers for each image-report pair. The raw radiology reports are read into a table, and the table column is used as the source for entity extraction.

The records are filtered by keeping valid CheXpert training image paths, by removing the non-empty raw reports, and removal of duplicates and unusable entries. The objective is to create a high-quality candidate pool where each record has enough clinical text for disease entity extraction for appropriate NER and VQA pipeline. The filtered data frame is converted into a list of clean record dictionaries and these records form a direct input to the Qwen-based entity extraction stage. Qwen2.5-14B-Instruct/4b is used as clinical entity extractor, operating over a fixed canonical vocabulary of disease/pathology terms.

The model is prompted to extract entities under a strict positive-only assertion rule extraction is performed one report at a time to prevent cross-report information leakage in the model's output. To ensure the extracted corpus is clinically informative, we retain only reports yielding more than a minimum threshold of positive entities ($>= 2$), positive image-report pairs.

Clinical Entity Extraction and Normalization: We use the pre-trained Qwen2.5/14B/Instruct model as a clinical entity extractor. The model is prompted to extract entities under a strict positive-only assertion rule to prevent cross-report information leakage in the model's output.

To ensure the extracted corpus is clinically informative, we retain only reports yielding more than a minimum threshold of positive entities (empirically set to 2), targeting a curated set of positive rich image report pairs. A controlled sanity checks are done to validate extraction behavior (true positives, correctly ignored negations, and handling of clinically supported uncertainty) prior to full-scale extraction.

Raw extracted entities are normalized to merge lexical variants of the same clinical concept. For instance, tumor, tumour, carcinoma is merged into a single canonical label, for a consistent final vocabulary. This normalization is performed to prevent VQA generation from rerunning and reinterpreting the entity labels.

Deterministic VQA Pair Generation

Given the finalized, normalized entity set per report, we construct binary VQA pairs using pure set-logic, with no LLM involvement at this stage:

Positive pairs: for every entity present the entity list, we generate a question of the form *"Is there $< entity >$ in this X-ray?"* with answer *Yes* and assign a *No* to negative pairs. The figure 2 gives the VQA pair generation.

To prevent semantically ambiguous negatives, semantic-sibling exclusion mechanism is done: entities that are clinically near-synonymous with a present finding *(e.g., pleural effusion and loculated effusion; cardiomegaly and cardiac enlargement)* are excluded from the negative candidate pool for that image. Rather than penalizing minor differences in clinically equivalent terminology, this approach focuses on the underlying clinical content, resulting in a fairer and less ambiguous benchmark.

Each generated pair is assigned a stable identifier linking it back to its source image and patient. The full positive/negative pool is shuffled before being split and saved. A final validation pass confirms field completeness, ID uniqueness, per-image Yes/No balance, and that no LLM-derived free-text answers leaked into the final dataset.

Domain Adaptation (llava_s)

Starting from LLaVA 1.5-7B, we perform supervised finetuning (SFT) on approximately $100{,}000$ (image, report) pairs from CheXpert Plus exposing the model to the radiological language. Finetuning uses Low-Rank Adaptation (LoRA) for one epoch at learning rate 5×10^{-5} with a cosine schedule and 0.03 warmup, updating only the injected low-rank adapters while the backbone stays frozen, using NVIDIA A100-SXM4-40GB. The resulting model, `llava_s`, understands single chest radiographs (remains single image) and cannot yet exploit reference cases. The Evaluation is performed on a held-out split on the curated VQA dataset as shown in table 2-4 in the ablation studies.

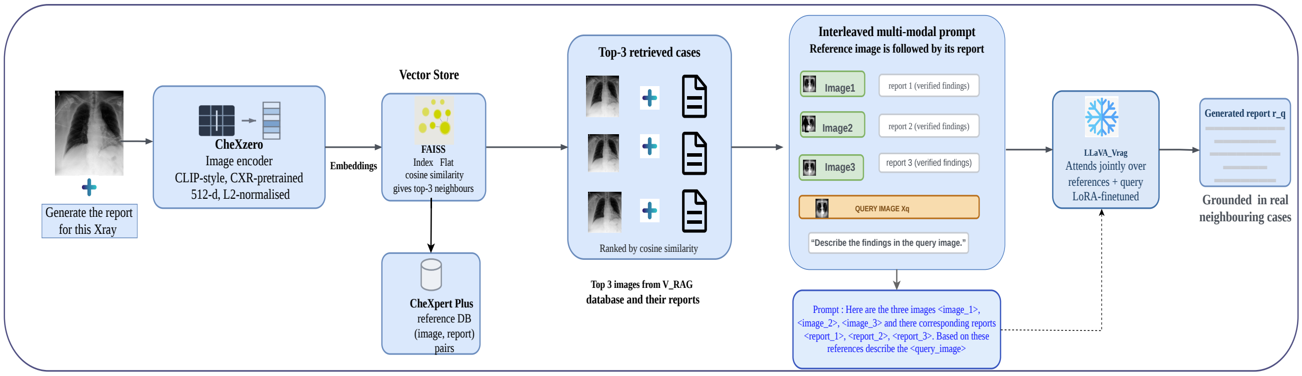

Figure 1: The V-RAG inference pipeline. The query radiograph is encoded with CheXzero into a 512-d vector and matched against a FAISS index built from the CheXpert Plus reference database; the top-3 image–report cases are placed in an interleaved multi-modal prompt (each reference image followed by its verified report, with the query image placed last) and fed to `llava_vrag`, which generates a report grounded in real neighbouring cases.

Inference using VRAG

Retrieval is performed in an image embedding space. Each database image is encoded with CheXzero (Tiu et al., 2022), a CLIP-style model contrastively pretrained on chest radiographs, yielding a 512-dimensional L_2-normalised vector. Vectors are indexed with FAISS using an inner-product index, so that inner product equals cosine similarity on normalised vectors. At query time the query image is encoded with the same encoder and the $k = 3$ nearest neighbours are retrieved; a self-leakage guard removes any neighbour that shares the query's patient identifier or image key, so a case is never retrieved as its own reference.

We also select CheXzero by an encoder comparison that measures *report-level* retrieval quality: for each query we retrieve by image similarity and then measure the cosine similarity between the query's report and the retrieved neighbour's report, where higher indicates a clinically more similar case. Relative to a random floor and an oracle that retrieves the most report-similar neighbour, CheXzero recovers more of the available headroom than the alternative encoder we test, yet even the best encoder closes only $\approx 8\%$ of the gap to the oracle, identifying retrieval quality as a standalone bottleneck and a direction for future work.

Stage 3: V-RAG Finetuning (`llava_vrag`)

We finetune `llava_s` on the mixed 33k dataset with LoRA for one epoch (same optimiser settings), updating only the adapters and distributing training across two GPUs. The resulting `llava_vrag` can attend across several images and their reports, closing the multi-image gap. At inference the full system encodes the query with CheXzero, retrieves the top-3 cases, and feeds their images and reports with the query image (placed last) to `llava_vrag`.

The `test_vqa` Hallucination Benchmark

Construction

Because scoring free-text hallucination directly is unreliable, we probe specific findings with a balanced binary protocol in the POPE style (Li et al., 2023). Each item pairs an image with a candidate finding (e.g. "consolidation") and asks whether the patient has it, with ground truth taken from the report. The construction pipeline (Figure 2) is fully automatic: positive findings are extracted from each report by a large language model (Qwen2.5-14B (Qwen Team, 2024)) performing constrained clinical named-entity recognition against a fixed canonical finding list, with explicit assertion and negation handling so negated and history-only findings are not extracted; a second model pass normalises the raw entities to the canonical vocabulary. For each retained image, positive questions are generated from its present findings, and an equal number of negative questions are sampled from findings absent in the report, *excluding semantic siblings* of present findings so a negative is never a paraphrase of a finding the patient actually has. This yields a per-image-balanced, sibling-safe probing set. We identically construct three benchmarks: CheXpert (6,940 questions / 1,000 images), MIMIC-CXR (7,088 / 1,000) and IU X-Ray (3,076 / 571), each $50/50$ balanced.

Metrics and Retrieval Conditions

Under this protocol a false "Yes" is exactly a hallucination (a finding claimed but not present) and a false "No" is a missed finding, so precision approximates one minus the hallucination rate, and the Matthews Correlation Coefficient (MCC) provides a robust summary of agreement with the truth. We evaluate all three models under three retrieval conditions: (i) **off**, query image only; (ii) **vrag**, the top-3 CheXzero/-FAISS references; and (iii) **random**, three references drawn at random from the database (additional context of no particular relevance). Comparing **vrag** against **random** isolates the contribution of retrieval relevance from that of extra context. For each of the nine configurations we report Accuracy, Precision, Recall, F1 and MCC; ROUGE is reported for completeness but, on single-word Yes/No answers.

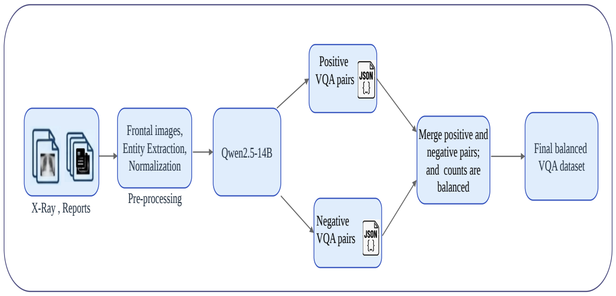

Figure 2: Automatic construction of the `test_vqa` probing benchmark (CheXpert Plus; MIMIC-CXR and IU X-Ray follow the identical pipeline). Reports are loaded and filtered to frontal, non-empty, one-per-patient cases; an LLM extracts and normalises positive findings; positive and semantic-sibling-excluded negative Yes/No pairs are generated and balanced.

Reinforcement Learning: When to Retrieve with GRPO

The three stage model progression discussed in the methodology treats retrieval as an external switch like a configuration we set, not a choice the model makes. This is a limitation, because retrieval is not free. Encoding the query, searching the index and prepending three reference images inflates the visual token budget. Hence we ask whether the model can learn when retrieval is worth its cost, and optimize that decision with Group Relative Policy Optimization (GRPO) (Shao et al., 2024).

Models: Both the policy and the reference are LLaVA-1.5-7B with the V-RAG adapter. The backbone remains frozen throughout and only the LoRA parameters are updated, giving 47M trainable parameters.

Data: The corpus comprises $190{,}868$ chest radiographs from CheXpert Plus, each paired with a parquet record carrying patient identifier, study number and the full radiologist-written report. The retrieval index holds $63{,}509$ studies embedded with the CheXzero encoder and searched with FAISS returning the top $k{=}3$ neighbours for a given query.

We construct the RL dataset as follows. All patients appearing in the $1{,}000$-image `test_vqa` evaluation set are removed from the $63{,}509$ retrieval rows, so no question the model trains on concerns a patient it will later be evaluated on. From the remainder we sample $8{,}000$ query images and draw four questions per image; two with ground-truth answer `yes` and two with `no`. This yields a per-image balanced set of $32{,}000$ examples. Each example is stored as one JSONL record holding the query path, the queried entity, the ground-truth answer, and the identifiers of its three FAISS neighbours:

Table 1: Reward under Eq. 2 for the four combinations of answer correctness c and retrieval decision z.

Correct (c)	**Retrieved** (z)	**Reward** (R)
1	0	$+1$
1	1	$1-\lambda$
0	0	0
0	1	$-\lambda$


```
{
  "query_key": "patient00845/study1/view1_frontal.png",
  "entity": "Cardiomegaly",
  "gt": "yes",
  "cached_refs": [
    {"png_key": "patient10203/study2/view1_frontal.png",
     "report": "PORTABLE CHEST: mild cardiomegaly..."},
    {"png_key": "patient07188/study1/view1_frontal.png",
     "report": "AP CHEST: heart normal in size..."},
    {"png_key": "patient09033/study1/view1_frontal.png",
     "report": "PA CHEST: Stable cardiomegaly..."}
  ]
}
```


Rollout structure: The rollout consists of three stages. In *Stage 1* the policy receives the query image and the question, together with an instruction to decide before answering whether it needs to consult similar reference cases, and to reply with exactly one of `RETRIEVE` or `ANSWER NOW`. We parse this response into a binary indicator

$$z = \begin{cases} 1, & \text{response begins with } \texttt{RETRIEVE}, \\ 0, & \text{otherwise.} \end{cases} \qquad (1)$$

In *Stage 2* the policy answers. If $z{=}1$, it is re-prompted with the query image, the three retrieved radiographs and their associated reports, and answers *yes* or *no*. If $z{=}0$, it answers from the query image and the textual question alone. *Stage 3* parses the answer and compares it against the ground truth, giving a correctness indicator $c \in \{0, 1\}$.

Note that the fallback branch of Eq. 1 assigns $z{=}0$ to any malformed Stage-1 response. This makes `ANSWER NOW` the default under parsing failure.

Reward: The reward couples answer correctness to an explicit retrieval penalty $\lambda \geq 0$. The reward computation is as follows:

$$R = c - \lambda z \qquad (2)$$

The table 1 enumerates the four outcomes. It rewards a correct answer, charges a fixed toll for consulting the index, and is indifferent between `yes` and `no` errors. Setting $\lambda{=}0$ recovers pure accuracy maximization and serves as our control.

Group-relative advantage ($\hat{A}_g$)**:** For each sampled question we draw $G{=}3$ independent rollouts $\{o_1, o_2, o_3\}$ from the current policy and score each with Eq. 2, giving $\mathbf{R} = [R_1, R_2, R_3]$. GRPO dispenses with a learned value network and instead normalizes within the group:

$$\hat{A}_g = \frac{R_g - \text{mean}(\mathbf{R})}{\text{std}(\mathbf{R}) + \epsilon}, \qquad g = 1, \ldots, G, \qquad (3)$$

Algorithm 1 GRPO training over the retrieval decision

Require: RL dataset $\mathcal{D}$; policy π_θ; frozen reference π_{ref}; penalty λ; group size $G{=}3$

for $t = 1$ **to** 3000 **do**
  Sample $(x, q, y) \sim \mathcal{D}$ with replacement
  for $g = 1$ **to** G **do**
    $z_g \leftarrow \textsc{Decide}(\pi_\theta, x, q)$ ▷ Stage 1, Eq. 1
    if $z_g = 1$ **then**
      $\mathcal{K} \leftarrow \textsc{FaissTopK}(x, k{=}3)$
      $\hat{y}_g \leftarrow \textsc{Answer}(\pi_\theta, x, q, \mathcal{K})$
    else
      $\hat{y}_g \leftarrow \textsc{Answer}(\pi_\theta, x, q)$
    end if
    $c_g \leftarrow \mathbb{1}[\hat{y}_g = y]$ ▷ Stage 3
    $R_g \leftarrow c_g - \lambda z_g$
  end for
  $\hat{A}_g \leftarrow (R_g - \text{mean}(\mathbf{R}))/(\text{std}(\mathbf{R}) + \epsilon)$
  $\theta \leftarrow \theta + \eta \nabla_\theta \mathcal{J}(\theta)$ ▷ Eq. 4
  if $t \bmod 20 = 0$ **then**
    log metrics
  end if
  if $t \bmod 200 = 0$ **then**
    save checkpoint
  end if
end for

with ϵ a small constant guarding the degenerate case in which all G rollouts receive identical reward. That case is common here: the action space is small enough that three rollouts frequently agree, in which case $\hat{A}_g{=}0$ for all g and the step contributes no gradient.

Objective: The sequence-level importance ratio is chosen as in (Zheng et al., 2025) and is given by : $\varrho_g = \pi_\theta(o_g \mid x, q)/\pi_{\theta_{\text{old}}}(o_g \mid x, q)$. The policy is updated by maximising

$$\mathcal{J}(\theta) \;=\; \mathbb{E}\left[\frac{1}{G}\sum_{g=1}^{G}\min\Big(\varrho_g \hat{A}_g,\ \text{clip}(\varrho_g, 1{-}\varepsilon, 1{+}\varepsilon)\,\hat{A}_g\Big)\right] - \beta\,\mathbb{D}_{\text{KL}}[\pi_\theta \,\|\, \pi_{\text{ref}}]\,, \tag{4}$$

where ε is the clipping threshold and β controls drift from the frozen reference. Only the LoRA parameters receive gradient, so the memory footprint matches that of Section 3. The training loop is as given by Algorithm 1 We sample one question per step with replacement, perform $G{=}3$ rollouts, and update. Metrics are logged every 20 steps and checkpoints written every 200, for 3,000 steps total. To trace the accuracy–retrieval trade-off we launch five runs in parallel with $\lambda \in \{0.00, 0.05, 0.10, 0.20, 0.30\}$, intending to recover a Pareto frontier over answer accuracy and retrieval rate.

Ablations

Setup

To validate the findings, all models are LoRA-adapted variants of LLaVA-1.5-7B loaded in 4-bit precision with the Unsloth runtime. Retrieval references are drawn from the fixed CheXpert Plus V-RAG database in all settings, so the external benchmarks measure cross-distribution *query* generalization with a common reference corpus. Inference is data parallel across two NVIDIA T4 GPUs. For the external corpora we exclude cases whose image is unavailable in the corresponding public collection; the evaluated counts are 6,940 (CheXpert), 7,056 (MIMIC-CXR) and 3,076 (IU X-Ray).

Main Results

The approach is evaluated using ROUGE, F1, Precision, Recall, and Matthews Correlation Coefficient (MCC) as evaluation metrics. Tables 2, 3 and 4 report all nine configurations on the three benchmarks. All the findings are consistent across the corpora.

V-RAG retrieval yields the best model. Across all three benchmarks, `llava_vrag` with V-RAG retrieval is the top-performing configuration: MCC $+0.472$ on CheXpert, $+0.458$ on MIMIC-CXR and $+0.346$ on IU X-Ray, with high precision (0.844, 0.850, 0.837 respectively) which implies few false-positive findings, the direct measure of hallucination.

Relevance, not context, drives the gain. For both trained models (`llava_s`, `llava_vrag`) the MCC ordering is monotone, off $<$ random $<$ vrag, on all three benchmarks (e.g. CheXpert `llava_vrag`: $+0.275 < +0.387 < +0.472$). Because **random** supplies the same amount of additional context as **vrag** but without relevance, the consistent **vrag**$>$**random** gap shows that the improvement comes from the clinical relevance of the retrieved cases, not from extra context alone. The base `llava`, which was never trained for multi-image reasoning, benefits from relevant retrieval but can be *hurt* by irrelevant context (on MIMIC-CXR and IU X-Ray its **random** MCC falls below **off**), underlining that exploiting references is a learned capability supplied by Stage 3.

V-RAG trades recall for a large drop in hallucination. Relative to the mode **off**, the **vrag** configuration of `llava_vrag` raises precision sharply (e.g. CheXpert $0.805 \rightarrow 0.844$; MIMIC $0.736 \rightarrow 0.850$) while recall decreases (e.g. CheXpert 0.541; IU 0.350): grounded in real neighbours, the model becomes more conservative and answers *"Yes"* only when the evidence supports it. Since a false *"Yes"* is the clinically dangerous error, this precision-favouring shift is the desired behaviour for hallucination reduction, though it comes at the cost of more missed findings.

Retrieval-Quality Analysis

The encoder comparison (Section Method) shows CheXzero recovers only $\approx 8\%$ of the report-similarity headroom to an oracle retriever, with the alternative encoder lower still. Since retrieval quality upper-bounds how useful the references can be, this identifies the retriever rather than the generator as the primary remaining bottleneck and motivates stronger CXR retrieval encoders as future work.

Table 2: CheXpert Plus `test_vqa` results (in-distribution, n=6940) set across three retrieval modes (*off*, *vrag*, *random*).Best MCC per model in bold.

Model	Mode	Acc	Prec	Rec	F1	MCC
`llava`	off	0.571	0.549	0.792	0.649	+0.159
	vrag	0.609	0.604	0.633	0.618	**+0.218**
	random	0.592	0.611	0.507	0.554	+0.187
`llava_s`	off	0.530	0.516	0.946	0.668	+0.108
	vrag	0.674	0.669	0.689	0.679	**+0.349**
	random	0.631	0.646	0.579	0.611	+0.263
`llava_vrag`	off	0.629	0.595	0.805	0.684	+0.275
	vrag	0.720	0.844	0.541	0.659	**+0.472**
	random	0.667	0.839	0.414	0.554	+0.387

Table 3: MIMIC-CXR `test_vqa` results (external, n=7056). set across three retrieval modes (*off*, *vrag*, *random*). Best MCC per model in bold.

Model	Mode	Acc	Prec	Rec	F1	MCC
`llava`	off	0.607	0.575	0.822	0.677	+0.238
	vrag	0.654	0.648	0.674	0.661	**+0.308**
	random	0.588	0.618	0.460	0.528	+0.182
`llava_s`	off	0.559	0.535	0.912	0.674	+0.167
	vrag	0.696	0.691	0.707	0.699	**+0.392**
	random	0.638	0.666	0.555	0.605	+0.280
`llava_vrag`	off	0.661	0.640	0.736	0.685	+0.325
	vrag	0.710	0.850	0.509	0.637	**+0.458**
	random	0.650	0.842	0.370	0.514	+0.363

Grpo Results and Retriever Collapse

Across all five settings of λ, the retrieval rate decayed to zero: the policy converged on emitting `ANSWER NOW` for every question and never consulted the index. The control run is what makes this informative. At λ=0.00 retrieval carries no penalty whatsoever, so a policy that found retrieved evidence useful would face no pressure to abandon it. Collapse under λ=0 therefore cannot be attributed to the cost term, and instead indicates that retrieved references were, on balance, degrading answer accuracy and the policy learned to avoid them because they were misleading, not because they were expensive.

This is consistent with the encoder analysis of CheXzero. The neighbours are selected by global image similarity, which need not align with the presence or absence of the specific entity a question asks about. Two radiographs can be close in embedding space while disagreeing on cardiomegaly. Under the fixed mode evaluation, the model has no choice but to condition on such neighbours; given the option to decline, it declines. We regard this less as a failure of GRPO than as a second, independent measurement of the retrieval bottleneck, arrived at by a different route.

We note two confounds that a stronger version of this experiment should remove. First, the malformed-response fallback in Eq. 1 makes non-retrieval the default, which biases the early policy toward z=0 before any learning occurs. Second, with G=3 and a four-valued reward, groups in which all rollouts agree produce zero advantage and no gradient; the effective number of informative updates is therefore well below 3,000. Whether retrieval collapse survives a larger group size, a symmetric parse fallback and an entity-aware retriever remains open.

Table 4: IU X-Ray `test_vqa` results (external, n=3076). set across three retrieval modes (*off*, *vrag*, *random*). Best MCC per model in bold.

Model	Mode	Acc	Prec	Rec	F1	MCC
`llava`	off	0.613	0.591	0.731	0.654	+0.232
	vrag	0.626	0.670	0.498	0.571	**+0.262**
	random	0.577	0.608	0.437	0.508	+0.161
`llava_s`	off	0.515	0.508	0.987	0.670	+0.089
	vrag	0.639	0.661	0.573	0.614	**+0.281**
	random	0.598	0.612	0.532	0.569	+0.197
`llava_vrag`	off	0.572	0.546	0.860	0.668	+0.176
	vrag	0.641	0.837	0.350	0.493	**+0.346**
	random	0.604	0.777	0.291	0.423	+0.266

Conclusion

Our method extends Visual Retrieval Augmented Generation on CheXpert Plus dataset with a LLaVA 1.5-7B backbone, and measures hallucination directly through `test_vqa`, a balanced probing benchmark. V-RAG fine-tuning with retrieval is the strongest configuration on all three corpora, reaching MCC +0.472 on CheXpert Plus and cutting the hallucination rate from 45.1% to 15.6%. The monotone ordering *off* < *random* < *vrag* shows the relevance of retrieval relevance. Two independent results, CheXzero closing only ≈ 8% of the oracle headroom and GRPO collapsing to zero retrieval even at λ=0, identify the retriever rather than the generator as the binding constraint. This limitation drives us to the future work to pursue reranking in the retriever and visual grounding of generated findings by cross alignment.

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